

# Payload Lift and Transport Using Decentralized Unmanned Aerial Vehicle Quadcopter Teams

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**Abstract**— This paper presents a decentralized and cooperative load lifting and transportation system using unmanned aerial vehicle quadcopters. The limitation of a single UAV to carry load is addressed in this study by creating a cooperative lifting system that can accommodate varying load weight. Cooperative, independent and scalable agents were implemented with decision making algorithm embedded in each agents. Decentralized sensing of load is done by the UAV and the group consensually decides if another UAV is needed to carry the load. The system can lift different weight by autonomously sending appropriate number of UAV depending on the load. Experiments were conducted to determine the responsiveness of the system in varying load weights. Experiment results showed that the developed system is robust and scalable.

**Keywords**—UAV, Decentralized Agents, Swarm Robotics.

## I. INTRODUCTION

Drone delivery systems are gaining popularity because of its convenience and practicality [1] [2]. There are several challenges to overcome to fully implement drone delivery systems. Such challenges includes maneuverability of load [3] [4] which is more problematic when size and weight of the payload increases [5]. One solution is to increase the number of vehicle that will carry the load [6] [7]. Furthermore, cooperative lifting, coordinated flight, and transportation control is an essential to successfully transport heavier load [8].

A solution is seen in [9] and [10] where load is cooperatively transported by more than one wheeled and fixed robot. The attempt to use two drones is seen in [11]. While in [12] displays the ability of a robotic swarm to control its centroid and maintain coordination in 3D space even if the number of robots is not constant. A swarm foraging algorithm is seen in [13] which enables a robotic swarm to maintain stability and consistency of its centroid while in translation in 3D space [14]. The use of quadcopters in the lifting of loads has eased manual labor. This also reduces the risk of accidents in dealing with hazardous materials that may risk worker safety. Instead of letting the workers

handle the toxic materials, a quadcopter may be used in their stead. In the accomplishment of carrying out a heavy load, multiple quadrotors can be used. This is done by controlling the UAVs through flight coordination and stabilization [15] [16].

In this context, this paper presents cooperative load lifting and transport utilizing autonomous decentralized and independent quadcopter drones. The robots autonomously decides if the current team members can lift the load. If the team failed to carry the load, another team member is called to help the team lift the load. The load is then transferred from the base location to the intended target area through the use of infrared cameras for navigation and detection.

This paper is organized as follows. Section II describes the movement of individual drones. Section III includes the analysis and discussion of the cooperative algorithm. Section IV illustrates the experiment procedures and its results. Section VI ends the paper with the findings and conclusion of the research.

## II. SYSTEM DYNAMICS

### A. Quadrotor Dynamics

Figure 1 shows the actuator diagram of a quadrotor drone. The configuration utilized is the + configuration. This configuration is oriented with one rotor placed in front and back, and two side rotors in left and right side of the robot. The rotation of each rotors are shown, adjacent rotors are rotated in opposite to maintain moment of the body to zero.

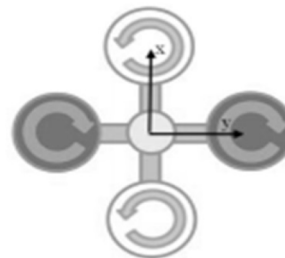
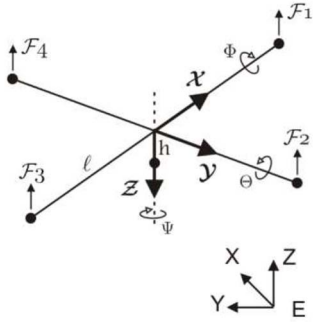


Figure 1. Quadrotor Actuator Diagram

Equation 1 shows the net moment of the robot body with respect to the moment contribution of the rotors.  $M_T$  is the total moment of the system, and  $M_i$  is the individual moment of the rotors.

$$M_T = \sum_{i=1}^4 M_i \quad (1)$$

Figure 2 describes the forces that acts and affects the orientation and movement of the quadrotor. As can be seen, rotations in three different axes are represented as  $\Theta$ ,  $\Phi$ ,  $\Psi$  or the pitch, roll and yaw angles.



**Figure 2. Forces and Moments Acting on the Quadrotor**

The figure also illustrates the upward force that the robot generates together with the gravitational pull or the weight of the robot. The positive net of the said forces is called thrust or lift. With different values of angles yaw, pitch and roll, the movement in the z-axis or altitude axis is converted into a lateral movement. Forward, backward, left and right movements are generated by creating an angle with respect to the plane of movement and increasing the thrust of the rotors.

The movement of the robot is modelled previously in [14] is described in the following equations:

$$\begin{aligned} I_{xx}\ddot{\Phi} &= \underline{\theta}\psi(I_{yy} - I_{zz} + \dots - h(\sum_{i=1}^4 H_{yi}) + (-1)^{i+1} \sum_{i=1}^4 R_{mxi}) \\ I_{yy}\ddot{\Theta} &= \underline{\Phi}\psi(I_{zz} - I_{xx} + \dots - h(\sum_{i=1}^4 H_{xi}) + (-1)^{i+1} \sum_{i=1}^4 R_{myi}) \\ I_{zz}\ddot{\Psi} &= \underline{\Phi}\psi(I_{xx} - I_{yy} + \dots - l(H_{x2} - H_{x4}) + l(-H_{y1} + H_{y3})) \\ m\ddot{z} &= mg - (c\psi c\Phi) \sum_{i=1}^4 T_i \\ m\ddot{x} &= (s\psi s\Phi + c\psi s\theta c\Phi) \sum_{i=1}^4 T_i - \sum_{i=1}^4 H_{xi} \\ m\ddot{y} &= (-c\psi s\Phi + s\psi s\theta c\Phi) \sum_{i=1}^4 T_i - \sum_{i=1}^4 H_{yi} \end{aligned}$$

The following three figures represent the forces exerted by the load on the on the drones in the different possible scenarios for load transport. Figure 3 corresponds to the minimum force or thrust drones 1, 2, and 3 must exert to lift the load,  $F_L$ .

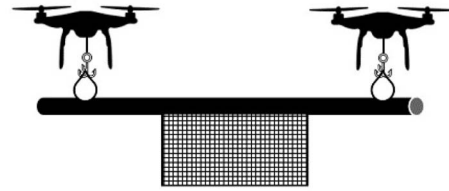


a) Graphical Illustration of Load Transport      b) Free-Body Diagram

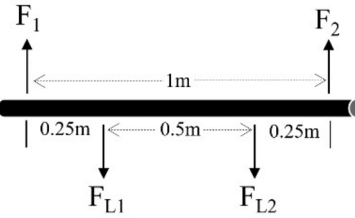
**Figure 3. Static Analyzation of a Single Load Transport**

$$\begin{aligned} \sum F_y &= 0 \\ \sum F_y &= F_L - F_1 \end{aligned} \quad (2)$$

In Figure 3, a single drone is tasked to carry the load. The drone must then counter all of the downward force caused the load.



a) Graphical Illustration of Dual Load Transport



b) Free-Body Diagram

**Figure 4. Static Analyzation of a Dual Load Transport**

$$\begin{aligned} \sum F_y &= 0 \\ \sum F_y &= F_{L1} + F_{L2} - F_1 - F_2 \\ \sum M_{L1} &= 0 \\ \sum M_{L1} &= -(F_1)(0.25) + (F_2)(0.75) \end{aligned} \quad (3)$$

(4)

$$\begin{aligned} \sum M_{L2} &= 0 \\ \sum M_{L2} &= -(F_1)(0.75) + (F_2)(0.25) \end{aligned}$$

(5)

In Figure 4, two drones are operated for load transport. The carrier with a pipe is used to hold the load. It is assumed that the downward forces are distributed between the points where the carrier connects with the pipe. With the further assumption that everything is asymmetrical, the two drones exert an

equal amount of force to lift the load. Eqns. 3, 4, and, 5 show the force calculations that each must provide.

### B. Quadrotor Team Control

Figure 5 shows the block diagram of the robot's control system. As can be seen, a layered architecture was implemented in controlling the robots. There are four layers designed in the control of the robot and the system. These layers are the behavior layer, trajectory control layer, attitude and altitude control layer, and the hardware layer.

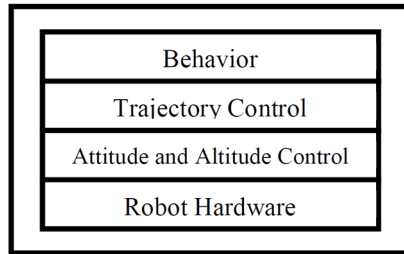


Figure 5. Quadrotor Control Block Diagram

The behavior layer is the highest layer of the system. It encompasses the autonomy in movement and movement decision making. Succeeding positions are planned in this layer. The placement or position of the robot with respect to the environment and nearby robots are considered in this layer. Once the next position is confirmed, the coordinates is then fed to the next layer which is the trajectory control layer. This layer plans the movement of the robot in the environment. Given the target position from the previous layer, this layer generates which path to traverse for a collision free movement. The attitude and altitude control layer generates the angles and thrust that the robot needs to traverse the planned trajectory. The robot hardware layer is the layer that translate all of the calculations into physical movements.

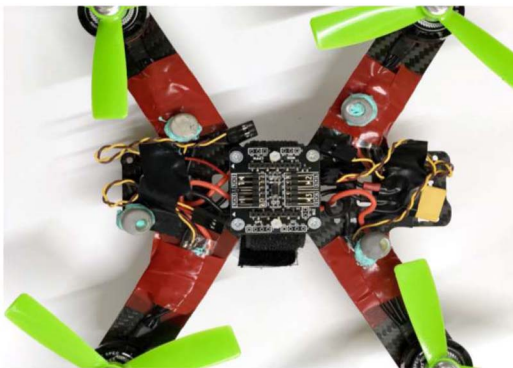


Figure 6. Actual Image of the Quadrotor

Figure 6 shows the actual quadrotor used in this study. The hardware layer and the processing elements are all located in a single physical device.

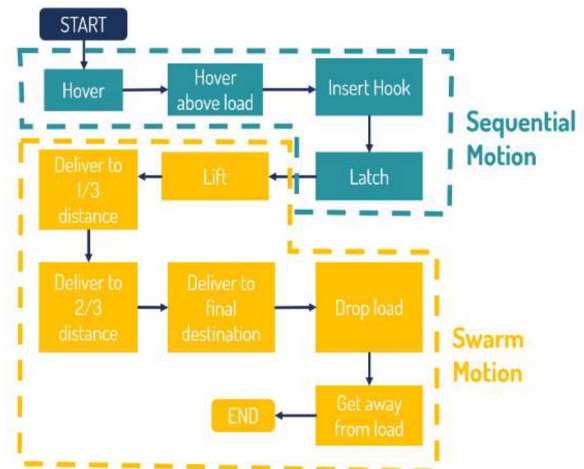


Figure 7. Flight path of the drone

Figure 7 shows the path that the drone takes to perform its tasks. Throughout this research, specified flight path are specified for the drone to follow. The sequential path is required for securing the hook to the carrier. The simultaneous path is required for the delivery of the load to its destination.

The swarm motion is the one responsible for instructing the quadcopters to lift and deliver the load to its target location. This function makes it possible for the drones to wait for each drone to arrive to its designated target location. This also makes it possible for the drones to always advance to the next target position simultaneously.

Table 1 shows the sequential motion of the drones is the initial movement of the drones. The program allows one quadcopter to initiate. The other drones should wait for the drone before them to properly latch to its hook before unlocking the next drone. Unlocking the drone would allow that drone to initiate and latch onto its intended hook. Once all the drones have properly latched, the swarm motion should begin.

TABLE I  
PSEUDOCODE FOR SEQUENTIAL MOTION

|                                                        |                         |
|--------------------------------------------------------|-------------------------|
| <b>Input (s):</b>                                      |                         |
| <i>Numquad</i>                                         | : number of quadcopters |
| <b>Output (s):</b>                                     |                         |
| Drone command.                                         |                         |
| 1: Send commands to CF1.                               |                         |
| 2: <b>for</b> i; from 0 to numquad                     |                         |
| 3: <b>if</b> <i>CF_status[i]</i> == 1 <b>then</b>      |                         |
| 4: <i>CF_lock[i+1]</i> = 0                             |                         |
| 5: <b>else if</b> <i>CF_status[i]</i> == 0 <b>then</b> |                         |
| 6:         Send commands to CF[i+1]                    |                         |
| 7: <b>end if</b>                                       |                         |
| 8: <b>end for</b>                                      |                         |

TABLE 2  
SINGLE LOAD TRANSPORT TEST RESULTS

| Trial      | Latching Time (s) | Unlatching Time (s) | Overall Transport Time (m's'') | Drop Accuracy |
|------------|-------------------|---------------------|--------------------------------|---------------|
| 1          | 29.97             | 13.45               | 1'52.08''                      | 78%           |
| 2          | 15.56             | 18.62               | 1'27.72''                      | 81%           |
| 3          | 39.61             | 3.75                | 1'40.75''                      | 88%           |
| 4          | 30.27             | 13.05               | 1'40.19''                      | 79%           |
| 5          | 32.35             | 20.35               | 1'33.28''                      | 81%           |
| 6          | 31.25             | 10.20               | 1'13.20''                      | 82%           |
| 7          | 29.63             | 13.42               | 1'22.03''                      | 74%           |
| 8          | 37.73             | 10.36               | 1'36.75''                      | 79%           |
| 9          | 32.51             | 24.23               | 1'11.65''                      | 80%           |
| 10         | 30.12             | 15.21               | 1'24.91''                      | 80%           |
| <b>AVE</b> | <b>30.79</b>      | <b>14.96</b>        | <b>1'26.31''</b>               | <b>80.2%</b>  |

The Failsafe function of the program ensures that the drone avoids unnecessary actions. This is also used to avoid operating the drone at maximum thrust. The first consideration for the Failsafe process is the amount of thrust that the quadcopter should produce. The maximum thrust determined that the drones can produce is 60,000. Due to this, the program only allows a maximum thrust of 57,000 so that it would minimize the damage that may occur to certain parts of the drone. The next consideration for this function is the maximum height that the drone can attain. Throughout the research, a height of 3.0 was used to execute the system. This is by reason of the program not allowing the drone to have a height greater than 4.0. If any of the aforementioned conditions is satisfied, the drones would automatically land. If none of it was achieved, the program would then instruct the quadcopters to perform the succeeding actions.

### III. EXPERIMENT RESULTS

The system is subjected to several setups which aims to determine its performance. A single robot test is done to determine the individual response of the robots with respect to the payload. The same setup is done while increasing the number of robots.

The experiment setup is segmented in three parts. The first step is by determining the position of the load and where the drone will attach. Second is the transport or trajectory planning and execution to where the load will be brought. Lastly the detaching stage where the load is safely landed and the drones will detach.

#### A. Single Load Transport

Several trials were conducted to determine the efficiency of the system. The results of each trials are summed and the average for each measured parameters were computed. The time for each stage and total time is also recorded and analyzed. Lastly, the accuracy rate of the overall performance is measured.

The amount of time elapsed for each segment of the load transport is measured. Furthermore, the total time to transport the payload is also measured. The latching time begins once the drone starts up. Table 2 shows the summary of results.

As can be seen, it took an average of 30.79s to perform the latching task. The timer for unlatching began once the drones drop the load. Results showed that it took an average of 14.96s to perform the unlatching task. Total time is the accumulative time from startup to landing. It took approximately 1'26.31'' to perform the entire test.

Figure 8 represents the flight path the drone took to deliver the load. As can be seen, the starting point of the drone is found at point (0,-6.5,-1.5) units. When the process of transportation commenced, the drone increased in altitude and moved towards the payload at point (0,-8.5,-1.5) units.

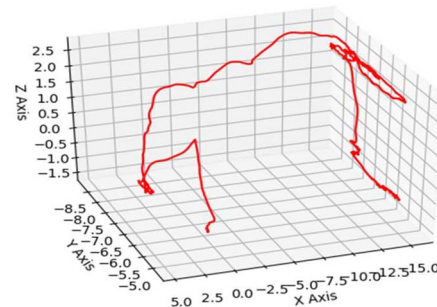
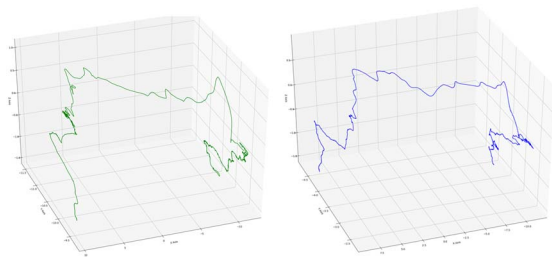


Figure 8. Trajectory of a Single Drone

#### B. Dual Load Transport

The test process for the latching of the dual load transport is similar to that of solo flight test, but with the addition of another drone performing the task. The time it took for each drone to accomplish its respective task was measured. Latching time took approximately less than 30 s for each drone. Unlatching time took 25.34 s. It took around 2'1.98'' for the drones to complete their task.



**Figure 9. Trajectory of Robot 1 and Robot 2**

#### IV. CONCLUSION AND RECOMMENDATIONS

Quadcopters working in unison are capable of transporting loads heavier than their individual capacity. Provided that the total load is less than the sum of their total capacity and is distributed evenly among each drone; the quadcopter can achieve stable flight. The PID can adapt to the increased load to an extent without causing wild movements in the drone.

This study deployed distributed processing of a decentralized agent system of flying robots for load lifting and transport. The cooperation of drones in lifting payloads is tested using simulation and actual experiments. The behavior emerged from the drones is cooperative and accurate in locating its position for the overall stability of the payload.

This work focused on this area due to the lack of initiatives present in the field. In order to compensate for that deficit, this research has designed an automated control of multiple quadcopters tasked with the transportation of a load to its designated target. This was done through the non-traditional means of employing autonomous flight and decision making.

During flight trials, it was observed that the drones, regardless of the size, can perform precise and exact movements under the right environment. The capacity of a single quadcopter can also be increased through the aid of other quadcopters done by cooperative means.

Performance can be improved by minimizing propwash and optimizing the weight distribution. Reducing the propwash of adjacent quadcopters lessens the air currents that push the drones away from one another yielding better drone stability. In a can result in a more stable flight. Drone formation and latching position affects the fraction of the load the drone in bearing. An optimal formation equally distributes the forces among the drones while keeping the carrier size to a minimum.

This indicates that the technological potential with drones possess the capability of reaching new heights that will be beneficial in opening new avenues for research and applications.

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